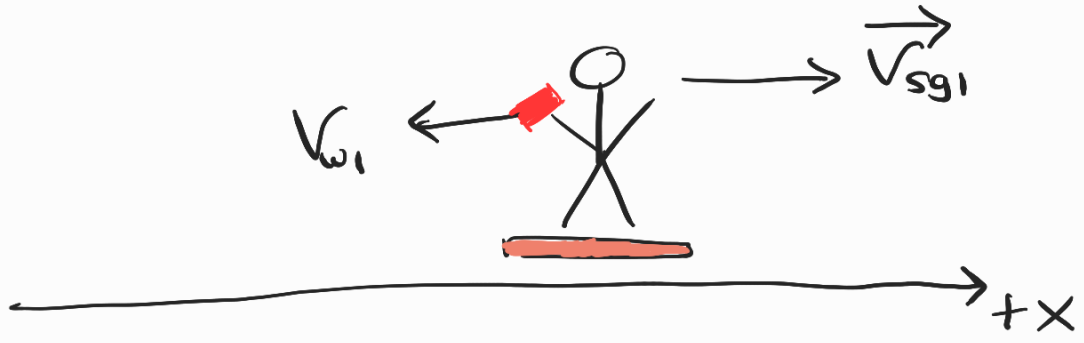
① $m_g = 40\,kg$ $\quad u_0 = 0\,m/s$

$m_s = 3\,kg$

$m_{w_1} = m_{w_2} = 5\,kg$



(a) $\vec{V}_{sg1} = ?$ $\quad u_{w1} = -7\,m/s$

$M = m_g + m_s = 43\,kg$

(m_g: αρκίδι, m_s: skateboard)

$\sum F_x = 0$

$\Rightarrow \vec{P}_0 = \vec{P}_1$

$\Rightarrow 0 = (M + m_{wk})\vec{V}_{sg1} + m_{w_1}\vec{V}_{w1}$ ①

όπου V_{sg1}, V_{w1} είναι οι ταχύτητες ως προς το έδαφος.

$u_{w1} = -7\,m/s$ → ταχύτητα βάρους ως προς το αρκίδι + skateboard

$\Rightarrow V_{w1} = u_{w1} + V_{sg1}$ ② → ταχύτητα βάρους ως προς το έδαφος

② → ①: $(M + m_w)\vec{V}_{sg1} + m_w(\vec{u}_{w1} + \vec{V}_{sg1}) = 0$

$\Rightarrow M V_{sg1} + m V_{sg1} + m_w u_{w1} + m_w V_{sg1} = 0$

$$\Rightarrow (M+2m_w)V_{sg1} = -m_w u_{w1}$$

$$\Rightarrow V_{sg1} = \frac{-m_w u_{w1}}{M+2m} = -\frac{(5)(-7)}{43+2(5)}$$

$$\Rightarrow \vec{V}_{sg1} = 0.66 \text{ m/s}$$

(b) $\vec{V}_{sg2} = ?$

$$\vec{P}_1 = \vec{P}_2$$

$$\Rightarrow (M+m_w)\vec{V}_{sg1} = M\vec{V}_{sg2} + m_w \cdot \vec{V}_{w2} \quad ①$$

$$V_{w2} = u_{w2} + V_{sg2} \quad ②$$

② → ① : $(M+m_w)\vec{V}_{sg1} = M\vec{V}_{sg2} + m_w(u_{w2} + V_{sg2})$

$$\Rightarrow M\vec{V}_{sg2} = (M+m_w)V_{sg1} - m_w u_{w2} - m_w V_{sg2}$$

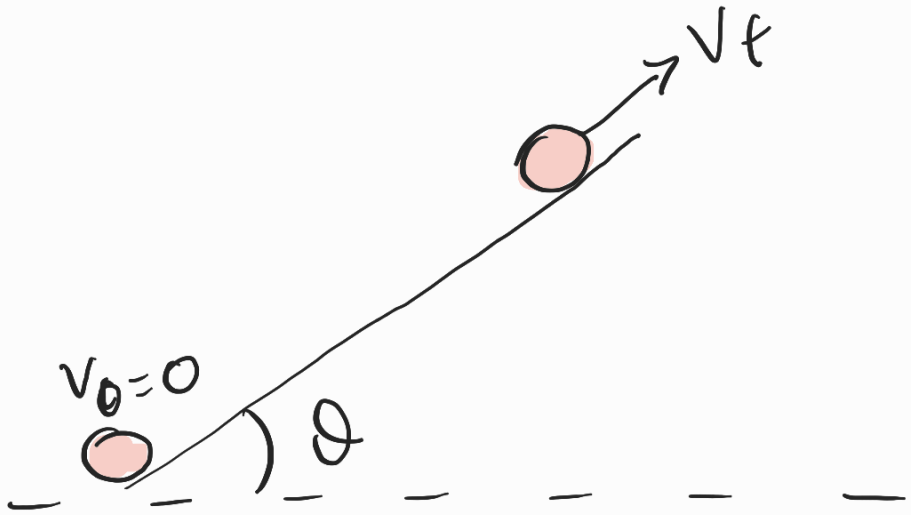
$$\Rightarrow (M+m_w)\vec{V}_{sg2} = (M+m_w)V_{sg1} - m_w u_{w2}$$

$$\Rightarrow \vec{V}_{sg2} = \frac{(M+m_w)\vec{V}_{sg1} - m_w u_{w2}}{M+m_w}$$

$$\Rightarrow \vec{V}_{sg2} = \vec{V}_{sg1} - \frac{m_w u_{w2}}{M+m_w}$$

$$\Rightarrow \vec{V}_{sg2} = (0.66) - \frac{(5)(-7)}{43+5} = 1.39 \text{ m/s}$$

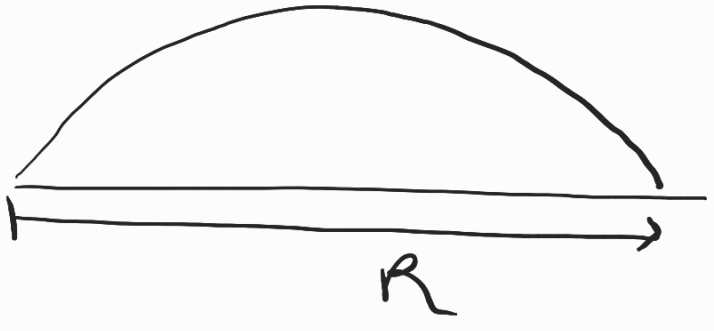
2



(a) $J = \vec{\Delta}_p = m(v_f - v_i)$

$\Rightarrow$ ωρέπει να βρούμε το $\vec{v_f}$

$R = 180m$



$$R = \frac{v_f^2}{g}\sin 2\theta \Rightarrow v_f = \sqrt{\frac{Rg}{\sin 2\theta}}$$

$$\Rightarrow v_f = \frac{180 \cdot 9.81}{\sin(2 \cdot 13)} = 65.2 \, m/s$$

$$\Rightarrow \underline{I} = (45 \times 10^{-3})(65.2) = 2.9 \, Ns$$

(b) $\Delta t = ?$

$$v_{av} = \frac{\Delta x}{\Delta t} \qquad v_{av} = \frac{v_0 + v_f}{2} = \frac{65.2}{2} \text{ m/s}$$

$$\Delta x = r$$

$$\Rightarrow \Delta t = \frac{2 \times 10^{-2}}{\frac{65.2}{2}} = 6.1 \times 10^{-4} \text{ s}$$

(c) $F_{av} = ?$

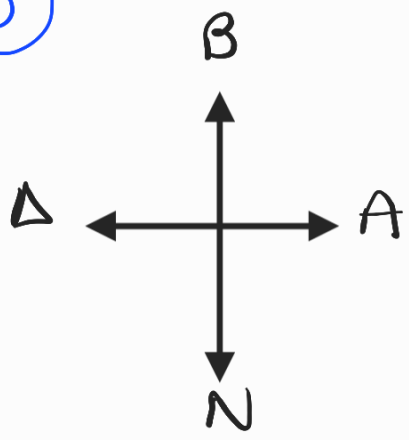
$$I = \Delta p = F_{av} \Delta t$$

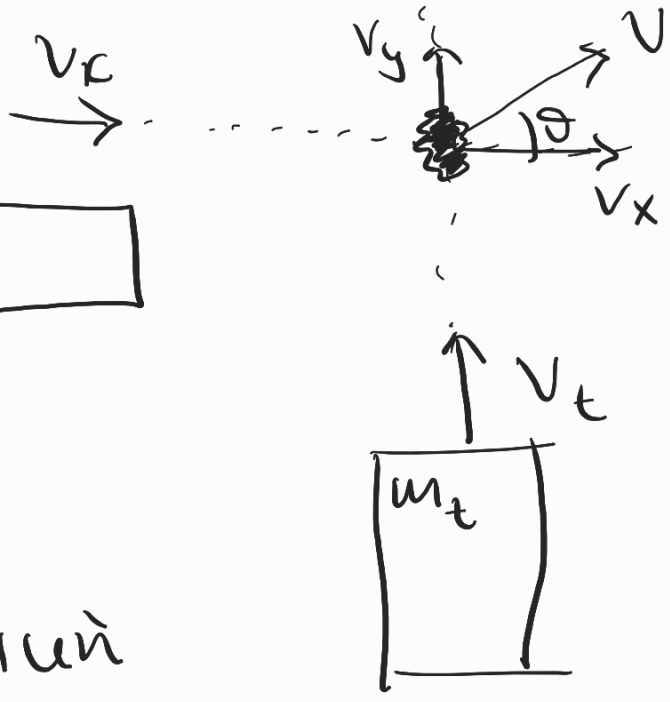
$$\Rightarrow F_{av} = I / \Delta t = \frac{2.93}{6.1 \times 10^{-4}}$$

$$\simeq 4.8 \times 10^{-3} \text{ N}$$

$$\Rightarrow F_{av} = 4.8 \text{ kN} //$$

3





Collision ⇒ Inelastic

$m_c = 1.2 \times 10^3$ kg $v_t = 50$ km/h

$m_t = 3 \times 10^3$ kg $\vartheta = 59°$

$v_t = 80$ km/h

$$P_i = P_f \Rightarrow m_c\vec{v_c} + m_t\vec{v_t} = (m_c + m_t)\vec{V}$$

X-axis : $m_c v_c + 0 = (m_c + m_t)\vec{v_x}$

$v_x = v\cos\vartheta \Rightarrow m_c v_c = (m_c + m_t)v\cos\vartheta$ ①

y-axis : $0 + m_t v_t = (m_c + m_t)v\sin\vartheta$ ②

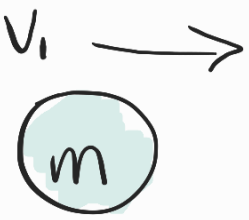
$$\frac{②}{①} : \frac{m_t v_t}{m_c v_c} = \frac{M \cancel{v} \sin\theta}{M \cancel{v} \cos\theta}$$

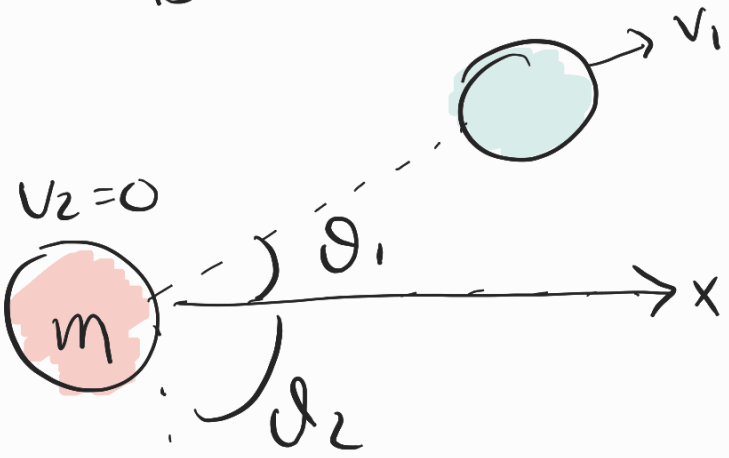
$$\Rightarrow v_c = \frac{m_t v_t}{m_c \tan\theta} = \frac{(3\times10^3)(50)}{1.2\times10^3 \tan 59}$$

$$\Rightarrow v_c = 75 \text{ km/h}$$

$$\Rightarrow v_c < v_{limit} \;//$$

(4) $m = 5\,kg$ $\quad v_1 = 2\,m/s$ $\quad v_2 = 0\,m/s$





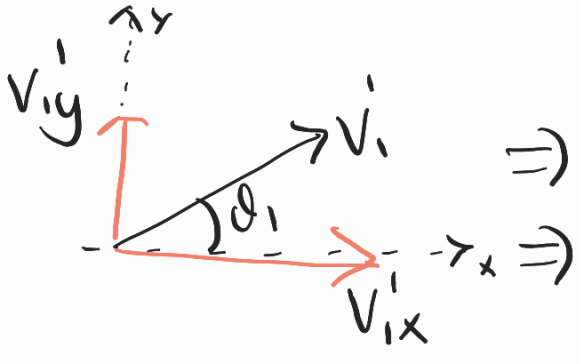
$\theta_1 = 30°$

$\theta_2 = 60°$



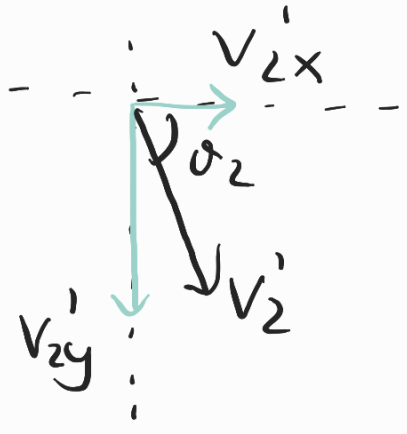
(a) $v_1' = ?$ $\quad v_2' = ?$

$$\vec{P}_i = \vec{P}_f \Rightarrow m\vec{v}_1 + 0 = m\vec{v}_1' + m\vec{v}_2'$$



$$\Rightarrow v_{1x}' = v_1' \cos\theta_1$$

$$\Rightarrow v_{1y}' = v_1' \sin\theta_1$$



$$\Rightarrow v_{2x}' = v_2' \cos\theta_2$$

$$v_{2y}' = -v_2 \sin\theta_2$$

x-axis: $P_x = P'_x$

$\Rightarrow m v_1 = m v_1' \cos 30 + m v_2' \cos 60$

$\Rightarrow v_1 = v_1' \cos 30 + v_2' \cos 60$ ①

y-axis: $P_y = P'_y$

$\Rightarrow 0 = m v_1' \sin 30 - m v_2' \sin 60$

$\Rightarrow v_1' \sin 30 - v_2' \sin 60 = 0$

$\Rightarrow v_1' = \dfrac{v_2' \sin 60}{\sin 30}$ ②

② → ① $v_1 = \dfrac{v_2' \sin 60 \cos 30}{\sin 30} + v_2' \cos 60$

0.5 (cos 60) 0.5 (sin 30)

$\Rightarrow v_1 = 2 v_2' \sin 60 \cos 30 + \frac{1}{2} v_2'$

$\Rightarrow 2 = (1.5) v_2' + (0.5) v_2'$

$\Rightarrow v_2' = 1 \text{ m/s}$

(2): $\Rightarrow v_1' = \dfrac{\sin 60}{\sin 30} = 1.73 \text{ m/s}$

(b) $K_i = \frac{1}{2} m v_1^2 = \frac{1}{2}(5)(2)^2 = 10 \text{J}$

$K_f = \frac{1}{2}(5)(1)^2 + \frac{1}{2}(5)(1.73)^2$

$= 2.5 + 7.5 = 10 \text{J}$

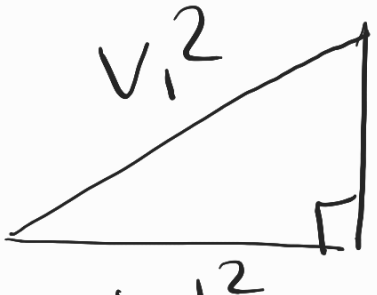
$\Rightarrow$ collision is elastic!

in

$K_i = K_f \Rightarrow \frac{1}{2} m v_1^2 = \frac{1}{2} m v_1'^2 + \frac{1}{2} m v_2'^2$

$\Rightarrow v_1^2 = v_1'^2 + v_2'^2$

$\Rightarrow$



$\Rightarrow 90°$ always!

(5)

(a) $\vec{A} = \vec{B} + \vec{C}$, $A^2 = B^2 + C^2 + 2\vec{B}\cdot\vec{C}$

$(\vec{B}+\vec{C})^2 = (\vec{B}+\vec{C})\cdot(\vec{B}+\vec{C})$

$= B^2 + C^2 + 2\vec{B}\cdot\vec{C}$ (2)

$\vec{A} = \vec{B}+\vec{C} \Rightarrow A^2 = |\vec{B}+\vec{C}|^2$

$\Rightarrow A^2 = (\vec{B}+\vec{C})\cdot(\vec{B}+\vec{C})$ (1)

$\Rightarrow$ (1) = (2) $\Rightarrow A^2 = B^2 + C^2 + 2\vec{B}\cdot\vec{C}$

(b) $\vec{P} = \vec{P_1} + \vec{P_2}$

$\Rightarrow \vec{P}\cdot\vec{P} = (\vec{P_1}+\vec{P_2})\cdot(\vec{P_1}+\vec{P_2})$

$\Rightarrow P^2 = P_1^2 + P_2^2 + 2\vec{P_1}\cdot\vec{P_2}$ (1)

ADE $\Rightarrow E_p = E_1 + E_2$

$E = P^2/2m \Rightarrow \frac{P_1^2}{2m} + \frac{P_2^2}{2m} = \frac{P^2}{2m}$

$\Rightarrow P_1^2 + P_2^2 = P^2/2m$ (2)

② − ①: $\vec{p}_1 \cdot \vec{p}_2 = p^2 - p^2$

$\Rightarrow \vec{p}_1 \cdot \vec{p}_2 = 0$

$\Rightarrow p_1 p_2 \cos\vartheta = 0$

$\Rightarrow \vartheta = 90^\circ$ //